

Module 08: Planning in High School

Learning Objectives

1. Define **Planning** within the context of High School.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In High School, we see Planning manifest in: * **Specific Example 1:** Decision trees in probability class are mathematical planning structures: at each branch point you evaluate the

expected value $E[X] = \sum(x_i * p_i)$ of each possible outcome, and the rational plan follows the path that maximizes expected reward (or equivalently minimizes expected free energy) -- for example, deciding whether to study for two tests by calculating which allocation of study hours minimizes your total expected grade loss. * **Specific Example 2:** The shortest-path algorithms you learn in discrete math (like Dijkstra's algorithm) formalize planning as mathematical optimization: the algorithm evaluates all possible routes through a weighted graph, maintains a priority queue of candidates ranked by cumulative cost, and greedily selects the path that minimizes total distance -- this is precisely how Active Inference agents plan by evaluating policies and selecting the one with lowest expected free energy across future time steps.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.